# Practical Work -7
# Application of ML Algorithms and NLP for Text Classification Task

**Due date: 09.04.2026**

Using any of the following benchmarked datasets

- Cyberbullying Classification:
  - https://www.kaggle.com/datasets/andrewmvd/cyberbullying-classification
  - https://www.kaggle.com/datasets/husawa23/hatespeech-dataset
  - https://www.kaggle.com/datasets/sayankr007/cyber-bullying-data-for-multi-label-classification

**Task:**

1. Data preprocessing:
   - Clean and normalize the raw text or features: removing noise from tweets or emails.
   - Implement appropriate encoding: TF-IDF for text or feature scaling for URL metadata.
   - Apply class balancing or augmentation techniques if your dataset is imbalanced and document the impact on your results.
2. Experimental design:
   - Establish a robust training/validation/test splitting protocol (70/15/15) to ensure unbiased evaluation.
   - Implement cross-validation strategies to verify the stability of your findings.
   - Identify and mitigate potential data leakage, ensuring that information from the test set does not influence the training or validation phases.
3. Model Training and optimization:
   - Select and train at least two supervised models (such as Random Forest, XGBoost); and two transformer models (such as LLMs & BERT, Llama or GPT4o).
   - Perform model optimization and hyperparameter tuning using grid or random search
   - Explore and justify your choices for activation functions, optimization algorithms, and regularization techniques to prevent overfitting
4. Performance Evaluation:
   - Calculate and report the following performance evaluation metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC.
   - Analyze the operational trade-offs between Recall (catching more threats) and False Positives (analyst alert fatigue).

**Note:** Summarize your findings in a report, include appropriate visualization plots to support your results on how the model handles malicious vs. benign intents, such as confusion matrix, ROC-curve, PR-AUC, t-SNE, etc.